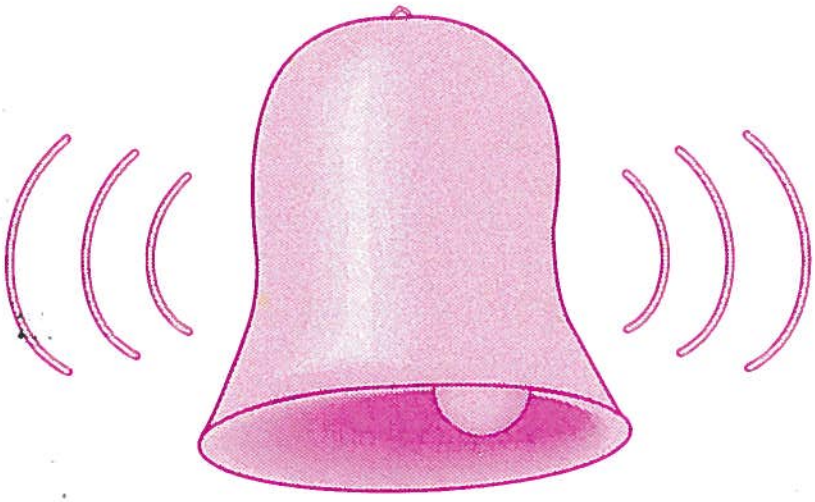
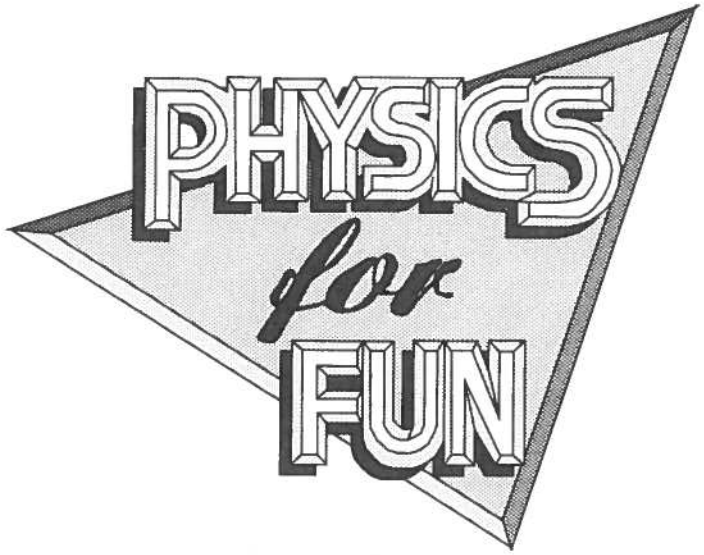
PHYSICS *for* FUN



SOUND UNIT

Science Discovery Program, University of Colorado at Boulder



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Overview of the *Physics for Fun* Curriculum

The *Physics for Fun* curriculum was developed by the directors of the Science Discovery Program, Physics professor John Taylor, and Engineering professor Kristina Johnson of the University of Colorado at Boulder. They worked together with a committee of public school teachers who piloted the lesson plans in their classrooms.

The goals of the curriculum are to provide students with an accurate base of knowledge in physical science and to increase student interest in science while doing it!

Each unit in the *Physics for Fun* series provides an activity guide which covers ten physical science topics and an accompanying videotape which reinforces the topics with high-interest demonstrations by the professors from the University of Colorado and by Denver's Channel 4 weatherman Larry Green.

Each topic begins by listing the concepts, objectives, and materials required. Every topic has at least one student activity. Some topics include optional and enrichment activities. Optional activities are more complex and/or time-consuming, and enrichment activities are designed for older or more capable students. A criterion-referenced test and scoring key for each unit are found in the Appendix.

The topic numbers in the videotape for each unit correspond to the topic numbers in the activity guide. In most cases, the video topics go beyond the activities, using technology that is unavailable to most teachers.

The *Physics for Fun* units are as follows:

- **Electricity**
- **Force and Motion**
- **Light**
- **Sound**

These activity-based units are designed for grades 4-6. The lessons can be adapted for older and younger students, however, and grades K-9 are encouraged to participate.

Introduction to the Sound Unit

The curriculum of this unit is organized around the following topics:

- **Topic 1: Sound Vibrations**
- **Topic 2: Sound Vibrations and How We Hear Them**
- **Topic 3: Sound Needs a Medium**
- **Topic 4: How Sound Travels**
- **Topic 5: Speed of Sound *(Optional)***
- **Topic 6: Stringed Instruments and Pitch**
- **Topic 7: Ear Harps**
- **Topic 8: Sound Words *(Optional)***
- **Topic 9: Wind Instruments**
- **Topic 10: Sound Sensations**

It is important to familiarize yourself with all the activities by trying them to determine which ones are most appropriate for your students. Some of the lessons may require more than one class period. Having students keep notebooks for recording their observations, experiences, and ideas will help students fully understand and integrate the concepts.

Most of the materials for this unit are easily obtainable. It is important to read through the list of materials far enough in advance to acquire what is needed. Some of the activities require students to bring in materials from home. Students should begin collecting the cans, combs, boxes, and toilet paper rolls needed for the topic "Sound Vibrations"; wire coat hangers for the topic "Sound Needs a Medium"; and the variety of bottles needed for the topics "Wind Instruments" and "Sound Sensations."

Accompanying Videotape

- **Topic 1: Sound Vibrations**

  A machine which is sensitive to pressure changes in air, called a soundscope, demonstrates that sound is a vibration.

- **Topic 2: More Vibrations**

  Sound vibrations change on a soundscope as sounds go from higher to lower and from louder to softer.

- **Topic 3: Sound Needs a Medium**

  Sound needs a medium like air to carry the vibrations. In outer space, explosions have no sound because there is no medium. A bell stops making its ringing sound as a vacuum is created around it.

- **Topic 4: How Sound Travels**

  Gliders traveling along an air track demonstrate how sound vibrations travel through the air.

- **Topic 5: Speed of Sound**

  Timing how long it takes sound to travel to a huge rock and then echo back demonstrates a means of measuring the speed of sound.

- **Topic 6: Stringed Instruments**

  Professor Taylor and Larry Green explain how various stringed instruments make and amplify sound.

- **Topic 7: Pitch**

  We learn what high and low pitches look like on a soundscope and what factors lead to different pitches and different octaves.

- **Topic 8: The Doppler Shift**

  A police siren speeding by illustrates the phenomenon of the Doppler shift. When sound vibrations are coming from an object which is moving toward us, the sound waves will be compressed and thus have a higher pitch. If the objects is moving away, the waves are elongated and the pitch is lower.

- **Topic 9: Wind Instruments**

  Professor Taylor and Larry Green demonstrate how various wind instruments function. All wind instruments consist of a tube with a hole in it—and one instrument has a tube which is thirty feet long!

- **Topic 10: Sound Sensations**

  A drum roll wraps up the learnings on sound and illustrates how the human ear works. The drum head vibrates when sound is near it, functioning like a human ear.

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Appendix

PHYSICS for FUN

Topic 1: SOUND VIBRATIONS

CONCEPTS

1. Sound is a tiny vibration, or movement, of the air.
2. Anything that makes a sound needs to vibrate in order to make the air vibrate.
3. Vibrations of the air cause tiny changes in air pressure.

STUDENT OBJECTIVES

1. Demonstrate in a variety of ways that objects must vibrate in order to produce a sound.
2. Build a soundscope, a device which shows visually that sound produces a vibration of the air.

MATERIALS

Students bring from home:

- One plastic comb
- A shoe box or other box that a rubber band can fit around
- Coffee can or other can with both lids removed (Only one per small group of students is essential, but it would be nice if each of the students made their own soundscope.)
- Empty paper towel or toilet paper roll (for the optional kazoo)

From the classroom:

- Hardcover textbook (one for every two students)
- Several packets of rubber bands, one packet being all the same size that will easily slip over the length of the textbook and the shoe box in such a way that there is some tension on the rubber band. Each child will use at least two rubber bands in this lesson.

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MATERIALS (continued)

- One box of waxed paper
- Three long pencils per pair of students
- Large balloons of good quality (plenty in case they tear)
- White glue
- Fragments of a mirror broken into pieces approximately 3 mm across (one for each soundscope)

NOTE: Students will need to blow up and tie off balloons a day or two before this lesson. This is a good way to create anticipation. Collect them in a large garbage bag and keep them until you begin this lesson. It would be a good idea to have a few extras.

CLASSROOM ACTIVITIES

Buzz It

This is a fun, although noisy, activity to kick off the unit on sound. It demonstrates that objects must vibrate to produce a sound. Actually, you should accommodate yourself to the fact that there will be a lot of sound produced in your classroom during the next two weeks.

PROCEDURE:

1. Give each student a piece of waxed paper large enough to cover his or her comb.
2. Challenge the students to use the comb and paper to make a sound.
3. In case they have trouble, they need to hold the waxed paper loosely against their combs, place their lips gently against the waxed paper and then hum into their combs. Students will need to experiment with different mouth and comb positions to get the buzz. The trick here is to be uninhibited and hum loudly.
4. Ask the students how this "instrument" enables them to make the buzzing sound: Do they feel the tickle on their lips? What is causing this? What happens if they hold the paper too tightly against the comb? Using a hand to stop the comb and waxed paper from vibrating, see if the students can understand what was vibrating and what happens when that vibration is obstructed.

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Explanation

Sound is produced by something vibrating. The waxed paper is vibrating against the comb producing the buzzing sound. When it is held tightly or a hand is held against it, neither paper nor comb can vibrate and the buzz disappears.

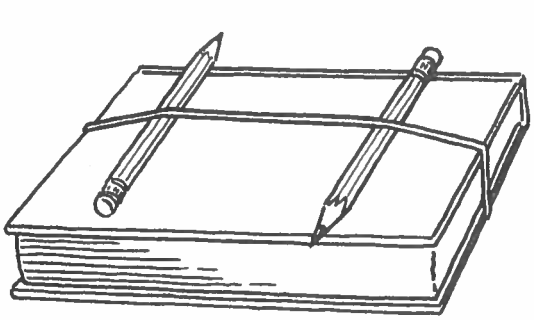
5. When students hum without the comb, sound is still produced. They may wonder what is vibrating to create this sound. Can they come up with an answer to this question? To explore further, have students place their hand on their vocal cords and turn to their neighbor and talk. See if they can feel their vocal cords vibrating.
6. The air passing through the vocal cords causes them to vibrate. Let the students see what happens when they hum with their mouths closed and then hold their noses.

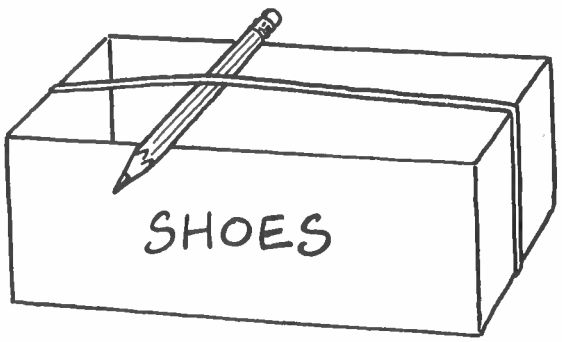
Rubber Band Serenade

Here students will study another vibrating object that produces a sound, a stretched rubber band.

PROCEDURE:

1. Have students work in pairs. Each pair should have one shoe box without the lid (the lid can also be used for the same purpose), one textbook with a hard cover, and two rubber bands.
2. Have students place a rubber band lengthwise around the book and another lengthwise around the box.
3. Students should then place two pencils under the rubber band on the textbook and one pencil under the rubber band on the box. These act as "bridges" and, in the case of the book, allow the rubber band to vibrate freely. A similar bridge is used in the violin and many stringed instruments.





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4. By plucking the rubber bands with their fingers, students can make a sound.

5. Ask the students what is producing the sound. What is vibrating? The students will easily see that the rubber band is vibrating. Is there anything else vibrating? By holding their hand gently next to the box while the rubber band is being plucked, most children will be able to feel that the box also is vibrating. Can the students think of something else that is vibrating?

Explanation

The rubber band is vibrating which causes the box to vibrate and that causes the air around it to vibrate. This also happens with the book, but the vibrations are too small to feel.

6. Holding the box or book on one's shoulder, up against the ear, will enhance the sound. Have students think about why this is so. The answer will be forthcoming with the topic "Sound Needs a Medium."

7. Allow the students a few minutes to experiment with their instruments and see what different sounds can be produced. Students may notice that moving the pencil causes the note to change.

Constructing a Soundscope

We have learned that to produce a sound, an object must vibrate. The vibrating object causes the air to vibrate. A sound is, in fact, a vibration of the air (or other medium). The soundscope is a simple device which enables us to observe that the air is vibrating. Students will make the soundscopes today and use them with the topic "Sound Vibrations and How We Hear Them." Students will use these in small groups, so only one per group is essential.

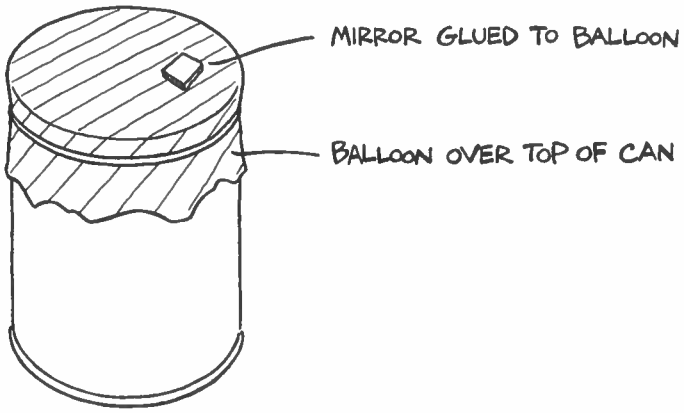
PROCEDURE:

1. Each student should have brought from home a can with both ends removed. Make sure the edges are not sharp. If they are, tape can be put around one rim to prevent cuts.

2. Hand out the inflated balloons, one to each student. Have students deflate the balloons either by untying them or by poking a small hole very near the knot. A pin or needle works best.

3. Have students cut off the knotted end of the balloon and slip the balloon over one open end of the can. Trim the balloon if necessary. Pull until the balloon is stretched firmly over the opening. If necessary, fasten the balloon securely with a rubber band.

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4. Have students place a dab of glue off center on the surface of the stretched balloon. Carefully place a fragment of broken mirror on the glue. For safety, the teacher should be the one to handle the mirror fragments and place them on the balloon surface. Set aside the cans in an upright position and allow them to dry for the next topic's lesson. This is the completed soundscope.

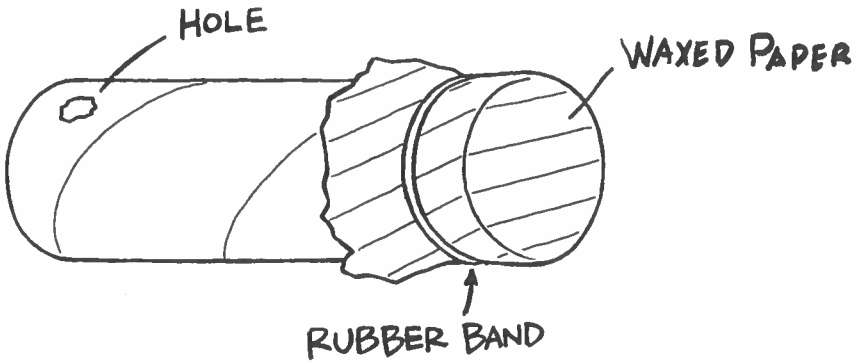
Kazoo Band

(Optional)

This is a fun activity involving vibration. You might encourage the music teacher to become involved, and the students can actually execute one of their choral pieces using their kazoos.

PROCEDURE:

1. Each student will have brought an empty paper towel or toilet paper tube to class.
2. Instruct each student to poke a pencil-sized hole through the tube, near one end.
3. Give each student a square piece of waxed paper to fit over the end of the tube.
4. Have students fasten the waxed paper over the end of the tube, using a rubber band to secure it. Be sure it is the end away from the hole.



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5. The kazoos are now completed. Humming into the open end creates the same buzzing sound as the comb and waxed paper. As before, students will need to experiment to get a good sound. If they place their fingers gently on the waxed paper they can feel it vibrating.

6. Conduct the students in a song. You can even do rounds or harmonies. Younger children might enjoy marching around the room as they play in the kazoo band.

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PHYSICS for FUN

Topic 2:

SOUND VIBRATIONS and HOW WE HEAR THEM

CONCEPTS

1. One way to detect sound vibrations is to use a rubber membrane (like the one in the soundscope made in the previous topic "Sound Vibrations"). You can see the membrane vibrating when there is a sound nearby, because the vibrations of the air push on the membrane and make it vibrate.
2. Air vibrating (that is, a sound) near your ear makes your eardrum vibrate. The eardrum is a small membrane of skin, quite like the rubber membrane in your soundscope. The vibrations of the eardrum are passed on (via three tiny bones and a liquid) to some hairs in the inner ear. The hairs are connected to nerves, which send a signal to your brain. When the brain receives this signal you hear the sound.
3. Having two ears makes it much easier to tell where a sound comes from.

STUDENT OBJECTIVES

1. Demonstrate with a soundscope that sound is a vibration of the air.
2. Learn the main parts of the ear and how they work.
3. Demonstrate that it is much easier to tell where a sound is coming from if you use both ears, rather than just one.

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Materials

- One battery-operated radio
- Some salt
- Two pieces of 2" X 4" scrap wood, each about 1' long
- Musical instruments from students who play them, or from you or a colleague
- The diagram of the human ear included in this lesson plan (p. 10)

Per group of three or four students:

- The soundscopes made for the previous topic "Sound Vibrations"
- One flashlight with good batteries
- Five books
- Any other sound sources the students can bring (whistles, rattles, kazoos, etc.)

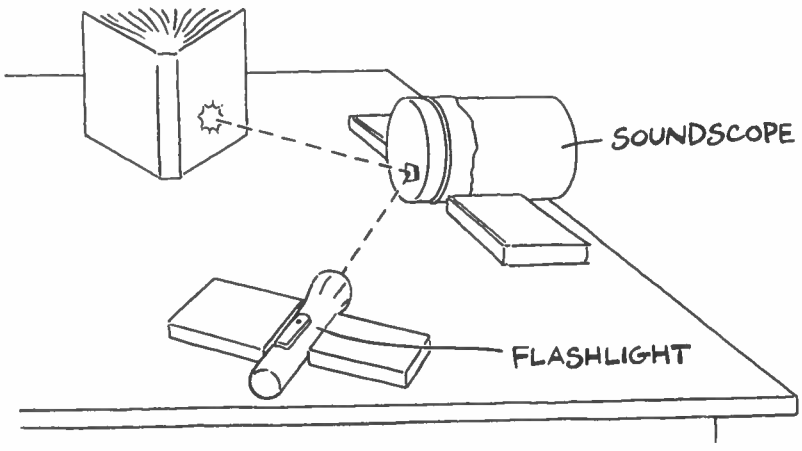
Classroom Activities

The Soundscope

The soundscope shows that sound is a vibration of the air. The vibrating air makes the rubber membrane vibrate visibly.

PROCEDURE:

1. Each group should place their soundscope at the edge of a table with its open end just off the edge. This allows them to bring a sound source close to the opening. Use two books to hold the scope steady.
2. Lay the flashlight on the table so that it points toward the mirror on the soundscope. Steady it with two books.
3. Stand a book on its end on the table as shown below.



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4. Adjust the positions of the scope, flashlight, and book until you get a clear patch of light on the book, reflected off the mirror, from the flashlight. It may help to rotate the soundscope to raise and lower the mirror, and to darken the room as much as possible.
5. Now try talking into the scope—quietly and then loudly. Observe the patch of light move when the membrane vibrates in response to a loud sound.
6. Try other sounds. Clap your hands. Whistle. Clap the scraps of wood. Try any musical instruments—high notes, low notes, loud and soft.
7. Try the portable radio. (Maybe do this as a demonstration for the whole class first.) When you turn the volume really high it may be possible to *see* the membrane itself vibrating.
8. For a grand finale, have the students gather around a table. Turn the radio on its back and place the soundscope on top of the speaker. Sprinkle some salt on the rubber membrane. Slowly turn up the volume and watch the salt dance.

Explanation

Any source of sound necessarily causes the air to vibrate, and the vibrating air causes the membrane to vibrate. The mirror and flashlight are just a way to make the vibrations of the membrane more visible. The louder the sound, the bigger the vibrations. Students may notice that the scope responds much better to some musical notes than others. This illustrates an interesting phenomenon called resonance: most things respond better to some kinds of vibration than to others, depending on their size and composition. For this reason, different groups' soundscopes may respond differently.

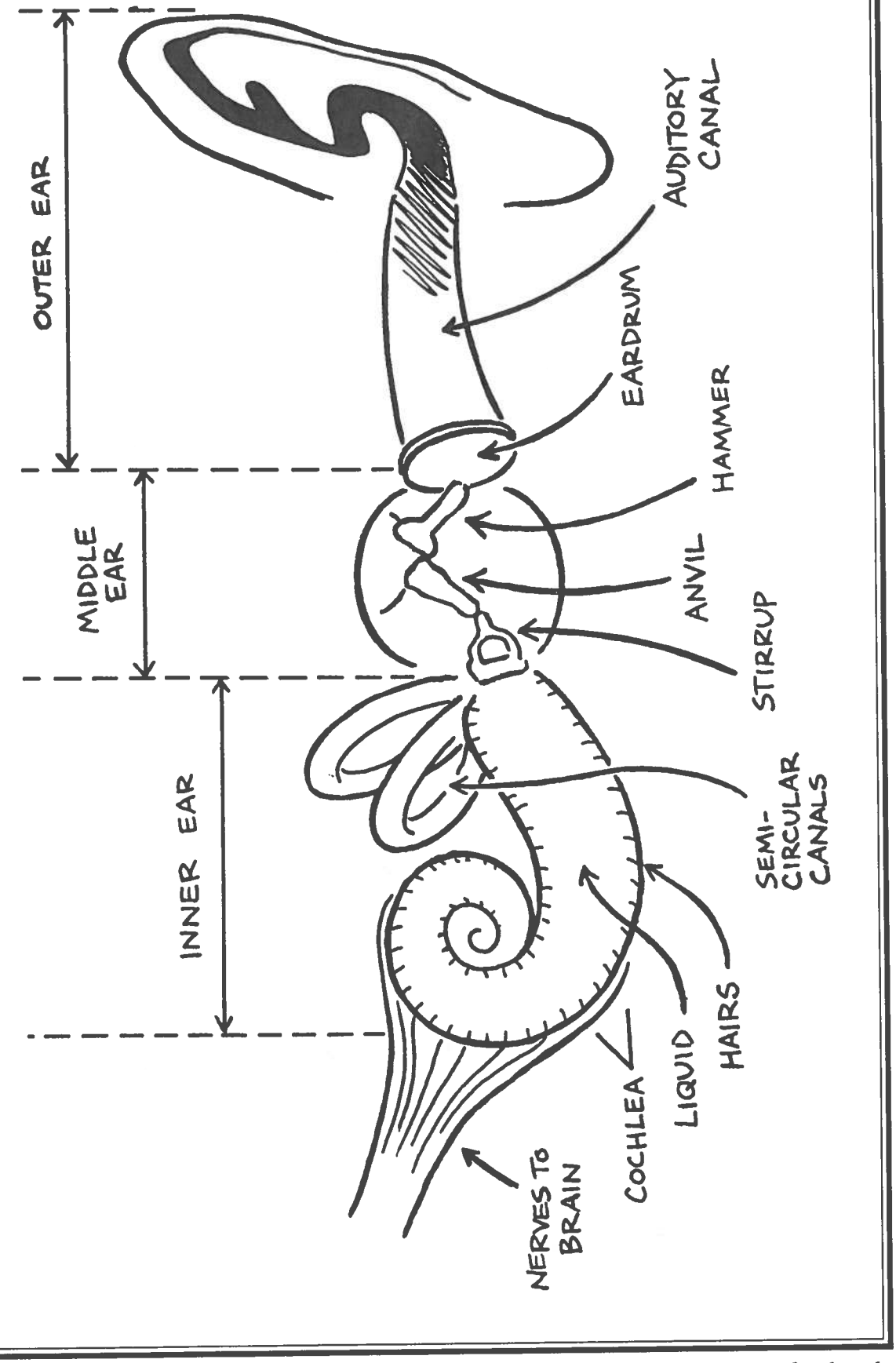
The Sensational Ear

PROCEDURE:

1. Reproduce the diagram of the ear (p. 10) on the board or an overhead projector, or distribute copies to the students. Discuss how the ear works in as much detail as you think is suitable. Students may want to find books and models to enhance this activity.

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Diagram of the Ear



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Explanation

Any reasonably loud sound makes all the air in the room vibrate. The horn-shaped part of your ear (the part we actually see) funnels the vibrations into your *auditory canal.* At the end of the auditory canal is the *eardrum,* which is very like the membrane on the soundscope. The vibrations of the air make the eardrum vibrate. The eardrum is connected to three tiny bones—the *hammer, anvil,* and *stirrup*—and makes these bones vibrate. The purpose of these bones is to amplify the vibrations. The last of the bones, the stirrup, is connected to the *inner ear* and makes it vibrate. The inner ear is filled with a liquid and is lined with tiny hairs. When the hairs vibrate they send a signal along nerves to the brain. When the brain receives that signal we hear the sound.

NOTE: The diagram also shows the semicircular canals that are part of the inner ear. These have nothing to do with hearing, but enable us to keep our balance.

Two Ears are Much Better Than One

(Optional)

This activity illustrates how having two ears helps us to tell where a sound is coming from.

PROCEDURE:

1. Arrange the class in a large circle and have a couple of volunteers sit at the center with their eyes closed or blindfolded.
2. Explain that you are going to point at one of the students in the circle, who will then make a sound by clapping hands. The volunteers in the center must try to point at the place where the sound came from.
3. Try this several times, allowing the volunteers to use both ears. They should be fairly successful at locating the source.
4. Now have the volunteers close one ear. It is important that the ear be truly closed. Repeat the experiment several more times. They will be much less successful at locating the source.
5. If there is time, repeat the whole experiment with different volunteers in the center.

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Explanation

Two ears help us locate the source in several ways: The ear that is closer to the source hears the sound a bit sooner, and our brain detects this tiny difference. Also the ear that is closer hears a sound which is a little louder, since the head blocks the sound to the other ear. This is similar to having two eyes for better depth perception.

6. Challenge the students to figure out why some animals like rabbits, deer, and wolves have two large, rotating ears. (Their large, rotating ears help them to hear in all directions for their prey or predators.)

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PHYSICS for FUN

Topic 3:

SOUND NEEDS A MEDIUM

CONCEPTS

1. The material that carries the vibrations of sound is called the *medium*. Without *any* medium there can be no sound. Therefore, sound cannot travel through a vacuum.
2. The usual medium for sound is air, but sound can also travel through wood, metal, water, and many other mediums.
3. Some mediums carry sound better than others.

STUDENT OBJECTIVES

1. Understand that no sounds are heard in outer space since there is no atmosphere, or medium, through which sound can travel.
2. Test different mediums to see how well they carry the vibrations of sound.
3. Construct hanger chimes and a string telephone and use them to observe how sound travels along string.

MATERIALS

- Transparency or copies of "Moon Concert" (included in this lesson plan, p. 15)
- Four or five wooden objects at least 6' long (table, piece of lumber, or wood floor)
- At least one long piece of metal (pipe, ladder, rebar, or playground equipment)
- Album or tape of whale sounds (optional)

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MATERIALS (continued)

Per pair of students:

- Cotton string (about 20' long)
- Piece of cardboard to wrap string around
- Two plastic cold drink cups (the flexible kind of plastic, and as large as possible)
- Two paper clips
- Nail for punching holes
- Two wire coat hangers
- Two 3' pieces of string or thread

CLASSROOM ACTIVITIES

What's Wrong With This Picture?

Students are challenged to discover why there are no sounds heard in outer space.

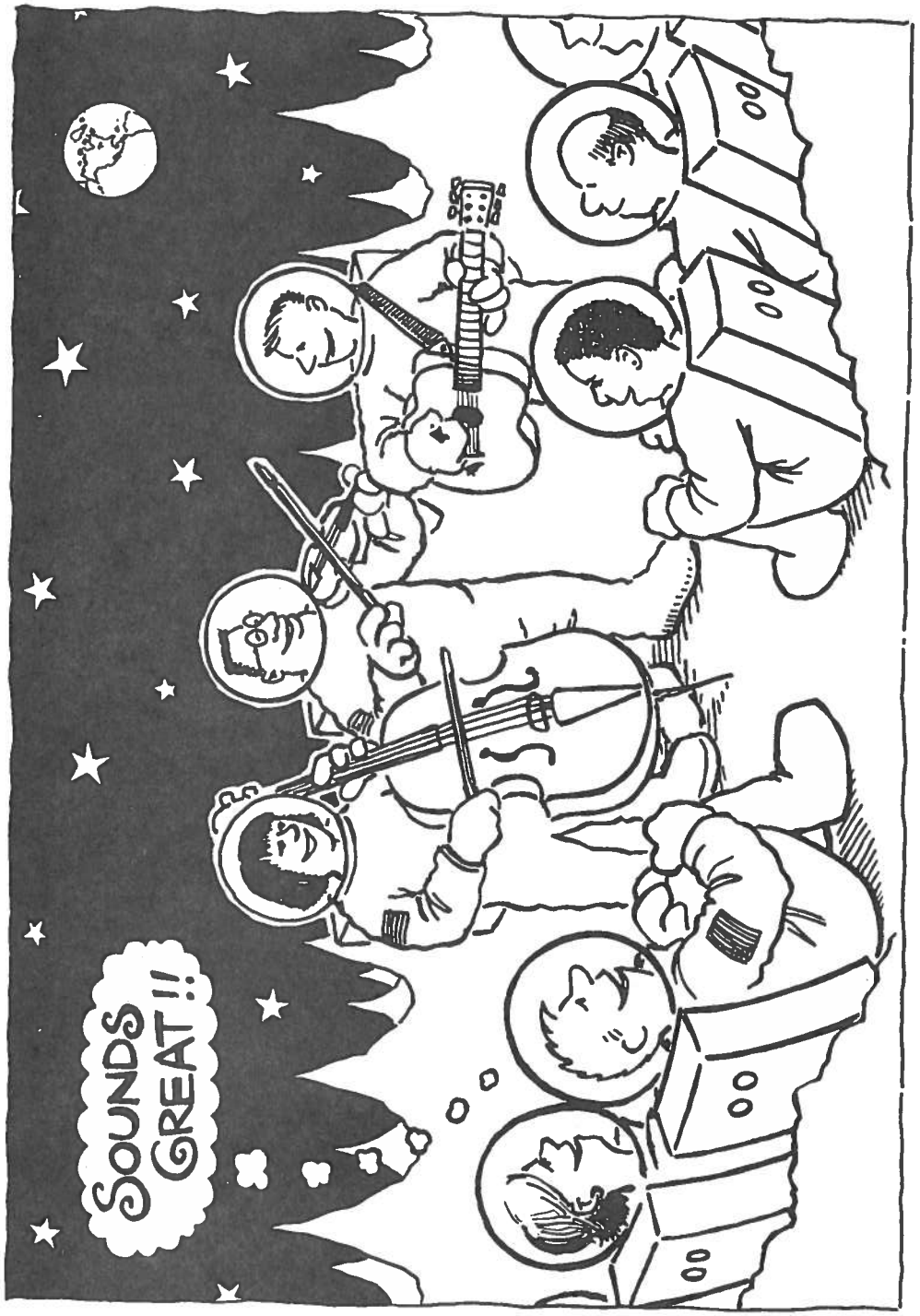
PROCEDURE:

1. Project a transparency or distribute copies of "Moon Concert" (p. 15)
2. Have each student write down what they think is the one major error in this picture.
3. Discuss students' responses with the entire group. Make sure they understand what a medium is and that a medium is necessary to carry sound.

Explanation

Sound needs something to carry it, like air, water or string. Whatever is carrying the sound is called the medium. Since there is no atmosphere in outer space (no medium), no sounds can be heard in space.

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Knock, Knock . . . Who's There?

Students will experience various mediums carrying sound.

PROCEDURE:

1. Have students gather around a long table.
2. Blindfold two or three students and have them stand at one end of the table. Tap very gently on the other end of the table so that the students cannot distinguish the number of taps. (It is easier to get an inaudible tap if you knock against the end of the table rather than the top.) Then have the students put their ears to the table. Tap again, just as gently. Ask if the students can hear you. Then tap several times and have the students tell you how many taps they heard this time. They will hear a much louder tapping sound with an ear to the table.
3. Ask students why they think the taps could only be heard, or heard more distinctly, with an ear to the table.

Explanation

There are several factors at work here: (1) The sound vibrations can be carried through the wood; (2) The table helps to focus the vibrations in the direction of the listener's ear; (3) When you knock on the wood, it is the wood that is set directly into vibration. The vibrations of the air are only a secondary effect. When the students put their ears to the table they bypass the need for the secondary and weaker vibrations of the air; and (4) When their ears are to the table, their ears are in direct contact with the vibrations of the table.

4. Now divide students into small groups, each group with a wood or metal object, and let each student have a chance to knock and then receive a "message." Make sure they tap first without the listener's ear to the object, so that they can notice the difference. Encourage them to try knocking and receiving with other mediums, such as carpet, concrete floors, etc. Which ones carry the sound of the gentle tapping?

NOTE: Students should be able to hear the tapping through wood and steel the best. They probably won't hear it through the concrete or rug, which tend to dissipate the vibrations.

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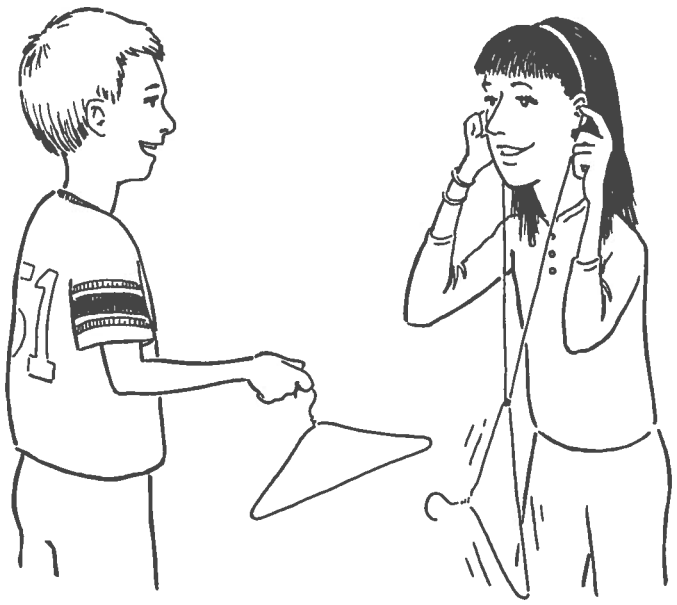
Hanger Chimes

(Optional)

Using the medium of string to carry the sound to their ears, students will hear the amazing sounds made by striking a hanger.

PROCEDURE:

1. In pairs, have students each loop a 3' length of string through their wire hanger. One student suspends a hanger by the strings while another student strikes it with a second hanger. Have them do this a few times to hear the feeble sound that is produced.
2. Next, have one student place the two ends of string on his/her index fingers and then place a finger in each ear. Have the other student strike the suspended hanger with the second hanger. The student with the strings in his/her ears will be aware of a very different sound than in Step 1. Strike the hanger on all sides and in different rhythms. Then let the other student be the "listener." Both students may want to experiment with banging their hangers against various objects in the room.



Explanation

The hanger makes the string vibrate more efficiently than it makes the air vibrate. The string carries the sound directly to the listener's ears.

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The String Connection

Students will also have the chance to talk through a medium, other than just air, and have their words be distinguishable. Students can work in pairs or groups to construct the string telephones.

NOTE: This is a fun activity, which could fill a whole class by itself. You might consider postponing it to use with either the topic "Speed of Sound" or "Sound Words."

PROCEDURE:

1. Each pair of students needs two plastic cups, approximately 20 feet of string, two paper clips, and a nail.
2. Have each pair of students punch a hole in the middle of the bottom end of each cup with the nail. Thread one end of the string up through the hole in the bottom and tie the thread to a paper clip. The clip will secure the string to the cup. Fasten the other end of the string to the second cup in the same way. This completes the string telephone.



3. Have the students take their telephones and find a space large enough to stretch the string out between them. Have the students take turns speaking in quiet voices into the cup, while another student receives the message. Challenge them to see how quietly they can speak into the cup and still be heard by the receiver.
4. Ask the students what is vibrating when the sounds travel to the receiver. Have one of the students hold a finger gently next to the string to feel it vibrating. The speaker may need to speak more loudly for the vibration to be felt.

Explanation

The vibrating vocal chords cause the air passing over them to vibrate. This causes the air near the mouth to vibrate, which in turn causes the cup and string to vibrate. This vibration travels through the string and vibrates the second cup, the air in that cup, and the listener's ear drum.

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Some students may ask how a string telephone is related to a real telephone. In fact there is a similarity. In a string telephone the vibrating air makes the string vibrate, and the vibrations travel along the string. In the real telephone the vibrating air causes a vibrating electric current to travel along a metal wire. At the receiving end the electrical vibrations recreate vibrations of the air.

5. Challenge students to add a party line to their phones and explain how it works. It is best to loop one telephone around the string of another. Students may also enjoy creating static on the line by running their fingers along the string.

Explanation

The vibration following along one string sets in motion everything it touches. Thus, the voice vibration traveling down the original string causes any additional string to vibrate in the same pattern and the voice is heard on the other end. Rubbing a finger along the string introduces extraneous vibrations.

6. If possible, have a 100' string telephone available for students to use in a hall or outdoors.

NOTE: If you have a tape of whale sounds, students can have the opportunity to hear sound traveling through water.

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PHYSICS for FUN

Topic 4:

HOW SOUND TRAVELS

CONCEPTS

1. When a sound source sends a sound out into a room, the air does not all start vibrating at once. Instead, the air next to the source starts vibrating first. This vibrating air pushes on the neighboring air and starts it vibrating. This process repeats over and over again, all the way out in every direction. In this way a listener anywhere in the room will receive the sound.
2. What travels from the source to the listener is the vibration, not the air itself.
3. When sound travels out in all directions it becomes weaker and weaker as it moves farther from its source.

STUDENT OBJECTIVES

1. Create an analogy of how sound travels.
2. Understand that when sound travels out in all directions it diminishes as it travels from its source.

MATERIALS

Students bring from home:

- Dominoes (as many as available)

From the classroom:

- More dominoes
- A metal slinky
- Transparency or copies of the "Spreading Sound" diagram included in this lesson plan (p. 25)
- A ball of dough made from flour and water or clay
- Yarn

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Classroom Activities

What's In a Clap?

Students will clap, making a sound that is just one quick vibration.

Procedure:

1. Review with the students that sound is started by something vibrating which, in turn, vibrates the air or other medium around it. The sound of the voice is a rather complicated sound with many different vibrations following one after the other through the air. In order to understand how sound travels, it is helpful to think about a simple sound that creates just one quick vibration.
2. Have the students clap their hands once. Pose the question, how do you think this sound is made? What is vibrating or moving? What is the medium through which the sound is traveling?

Explanation

Students know that sound is made when an object vibrates (moves back and forth). A clap is an example of just one quick movement (a simple vibration) rather than a series of vibrations, like human speech. When you clap your hands, you force the air between them outward. This air pushes the air next to it, and so on. In this way, the movement travels through the air to the listener.

The Domino Effect

The chain reaction of falling dominoes that are lined up is a good analogy for the way sound travels. Here students will create a domino chain reaction.

Procedure:

1. Have students work in groups so that each group has many dominoes.
2. Have the students line up the dominoes on a hard surface, standing them on end about 1/2" apart.
3. When they are all lined up, one student can gently push the first domino. Watch the chain reaction as it spreads down the line.

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4. Ask the students how this is similar to the way in which a clap (a single disturbance of the air) travels through the air.
5. You may want to let the students have some fun making a very long line of dominoes.

Explanation

The important point to recognize here is that it is the disturbance, or the energy, that is moving through the dominoes, *not* the dominoes themselves traveling to the "listener." This is an important concept because children often think that the air itself (or other medium) travels from the sound source to the listener, when it is really just the energy of the vibration that travels.

What Moves the Slinky?

Everyone has played with a slinky. Physicists also use slinkies as another tool to demonstrate how sound travels.

PROCEDURE:

1. Have three students come to the front of the room to demonstrate the slinky. While two students stretch the slinky, have the third tie pieces of colored yarn to a couple of the rings on the slinky.
2. Ask the students what will happen to the pieces of yarn (how they will move) when a vibration is sent through the slinky.
3. Have the two students who are holding the slinky stretch it across the room (not too tightly). The third student can then send a vibration through the slinky by pulling three or four rings together near one end and then letting them go. The vibration should move through the slinky. What happens to the yarn?

Explanation

The yarn will move slightly forward and backward as the energy of the vibration passes through the slinky. The same thing is happening to each ring of the slinky. This is what happens to the air (or other medium) as a sound vibration moves through it. Students may also notice that when the vibration reaches the end of the slinky it begins to come back again. This is also true of sound, which will bounce off an obstacle in its path, such as a wall, and travel in a new direction. This is called an echo.

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Spread the Dough

(Optional)

Through this demonstration, students will begin to understand why sound gets softer as it spreads out from its source.

PROCEDURE:

1. Show the students the picture ("Spreading Sound," p. 25) of a sound spreading out from a bell and being heard by the three people, A, B, and C. Ask them why they think the sound heard by C is much fainter than the sound heard by A. If they answer "because it is farther away," then the question becomes, "Why does being farther away make it softer?"

Explanation

As the sound travels outward in all directions it spreads out over a larger and larger area. The same amount of sound energy is being spread over a larger area, and the sound becomes softer. It may help to show the students the following analogy. (If you don't want to, you needn't actually do this; just describe it.)

2. Have a student push a ball of dough down on a table or round pizza pan so that you have a very thick, small pizza.
3. Now start spreading out the pizza dough and show how the more you spread it out, the thinner it becomes. This is analogous to sound. As the sound moves away from the source, the same amount of energy is spread out over a larger area and hence gets "thinner" (softer). This is why a person standing close to a bell will hear it better than a person on the other side of the room.

NOTE: For your information, this is called the Inverse Square Law. It states that the intensity of sound is inversely proportional to the square of the distance from the source. The formula is:

$$\textbf{Intensity} \propto 1/d^2$$

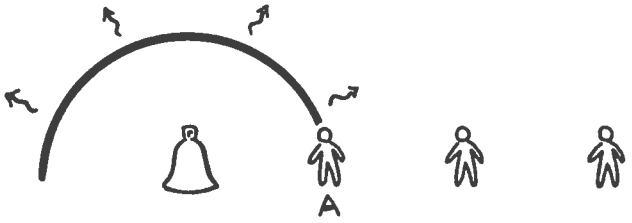
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Spreading Sound

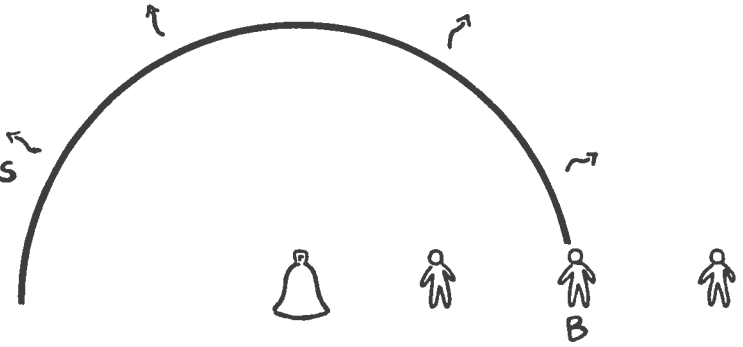
1. HAMMER HITS BELL



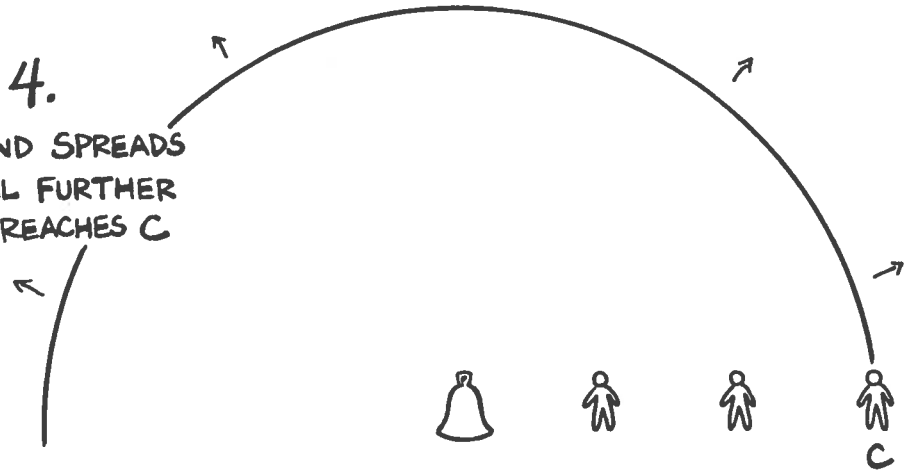
2. SOUND SPREADS OUT AND REACHES A



3. SOUND SPREADS FURTHER AND REACHES B



4. SOUND SPREADS STILL FURTHER AND REACHES C



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PHYSICS for FUN

Topic 5:

SPEED OF SOUND

(Optional)

Special Note: **This topic's second activity, "Measuring the Speed of Sound," is a challenging activity, probably suitable only for grades five and up, and better accomplished with a small group. If you decide to omit it, this would be an opportunity to catch up on anything missed in the past topics of this unit.**

CONCEPTS

1. When sound travels outward from a source, it travels at a speed of about 1,100 feet per second (or 340 meters per second). This means that it takes about 5 seconds for sound to travel a mile.
2. An echo is a reflection of sound off a wall or other surface. When you hear an echo of yourself, the sound has traveled from you, to the wall, and back to you.

STUDENT OBJECTIVES

1. Learn how to measure the distance of thunder and lightning.
2. Measure the speed of sound.

MATERIALS

For measuring the speed of sound:

- One long tape measure (If it is 50 feet or more, your job will be easier.)
- Two pieces of 2" X 4" scrap wood, each about 1' long
- At least one stop watch
- Calculators (may help with dividing decimals)
- A location with a good, clear echo as described on p. 29 (If your school is located in a noisy neighborhood or there is no suitable spot, this activity may be impractical.)

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Classroom Activities

Where's the Thunder?

Many of your students probably already know the old trick for figuring out how far away a clap of thunder is, but now they can understand how it works.

Explanation

Lightning and thunder occur at the same time in a thunderclap. The light from the lightning travels so fast (200,000 miles per second) that it takes almost no time to reach your eye. Thus you see the lightning at almost the same instant it occurs. On the other hand, the sound from the thunder travels more slowly, taking about 5 seconds to go each mile. Therefore you hear the thunder a bit later. If it arrives 5 seconds later, then the thunderclap was 1 mile away; if it arrives 10 seconds later, then it was 2 miles away, and so on. The general rule is this: Count how many seconds between the lightning and the thunder, divide this by 5 and you have the number of miles.

Procedure:

1. Explain what happens when we see lightning and hear thunder. Try to get them to figure out what the general rule is by going through some examples. (How far is it if you hear the thunder 5 seconds after you see the lightning? What if it is 10 seconds? What if it is 7 seconds? And so on.)
2. If they don't already know how, teach them how to count seconds without a watch. (One way is to count slowly: "1 potato, 2 potato, 3 potato, ...")

Measuring the Speed of Sound

Students will measure the speed of sound by timing an echo off a wall. It is important that you try this activity a few times before doing it with your students to know the technique and to know if the students' measurements are about right.

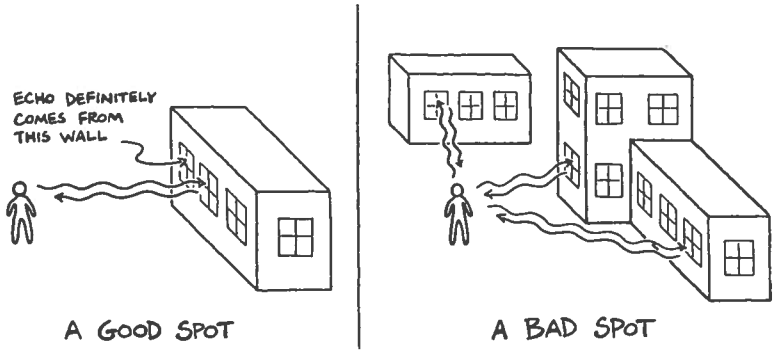
NOTE: This procedure is a little complicated because sound travels very fast, making its speed tricky to measure. Nevertheless, this is a fun and challenging activity.

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PROCEDURE:

1. Take your students to the spot (which you have already located) where you can get a good, clean echo off a wall of the school or a nearby building. This spot should be more than 100 feet from the wall that gives you the echo. It is important to have a wall that stands alone so that you know where the echo is coming from.

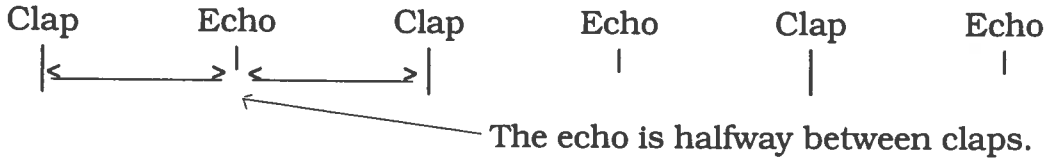
NOTE: If the wall is 100 feet away, the echo will return in about 1/5 of a second and is quite hard to hear distinctly. This is why it is important to find a wall that is as far away as conveniently possible.



2. Have the students measure the distance between the place where you are standing and the wall. Point out that the sound will travel *twice* this distance. Divide the students into small groups, as many groups as you have stop watches, but be sure there is an odd number of groups. Students who are not clapping or timing should act as recorders. They should record the distance from the clapping students to the wall and record the times indicated on their groups' stop watches.

3. Have students guess how long it would take for a sound to bounce off the wall and return to them (an echo!).

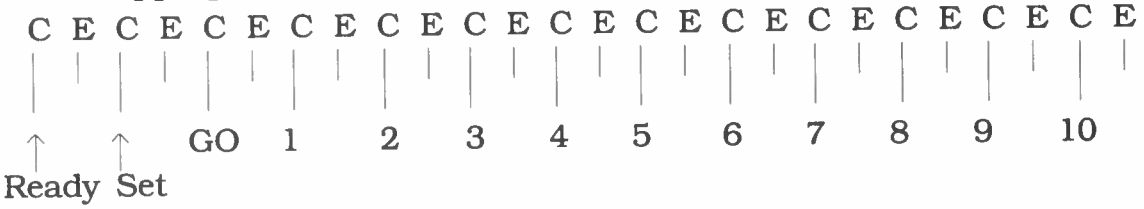
4. In order to get an echo, you need to make a sharp, loud sound. Clapping the two pieces of wood together works well. Clap the boards regularly to create a rhythm—clap-echo-clap-echo-clap-echo . . . . In this way, you can hear the echos clearly. Have the students practice this rhythmic clapping so that the times from "clap" to "echo" and from "echo" to "clap" are the same. This sounds hard, but you will find you can hear it quite easily.



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NOTE: If you could time the interval from "clap" to "echo" directly (one round trip), then all you would have to do is divide the distance traveled by the time taken, and you would have the speed of sound. Unfortunately the time of one round trip is far too short to measure with any accuracy. Therefore, we must use some cunning and measure the time for several round trips.

5. Select two or three students who will be the clappers. While one student claps, have the students with stop watches time ten complete intervals from "clap" to "clap." This takes a little care. You could help them by counting down: "ready, set, go" in rhythm with the clapping:



6. Have them repeat Step 5 several times, allowing different students to clap. Be sure all results are recorded by each group.

7. Return to the classroom to do the calculations.

8. Multiply the distance from the clapping student to the wall by two. This gives the round trip distance traveled by the sound.

9. Record each group's time for the first trial on the board. You must now choose "the best" time. A good way of doing this is to write the times in increasing order and pick the middle one. This is called the median.

10. Divide the chosen time by ten to find the time of one interval from "clap" to "clap." Next divide by two to get the time from "clap" to "echo."

11. Divide the round trip distance (Step 8) by the time interval (Step 10) and you have the speed of sound.

12. Here is a sample calculation:

First:
(distance from clapping student to the wall) = 145 feet
Therefore:
(round trip distance) = 145 x 2 = 290 feet
and
(time for ten intervals) = 5.5 seconds
Therefore:
(time of one interval) = 5.5 X 10 = 0.55 seconds
and
(time for one round trip) = 0.55 X 2 = 0.275 seconds

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Therefore:

(speed of sound) = (distance traveled) ÷ (time taken)
= (290 feet) ÷ (0.275 seconds)
= 1,055 feet per second

If you get an answer anywhere between 1,000 and 1,200 you have done well. Good luck!

13. Repeat calculations for the other time trials, either as a whole class or in the small groups.

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Topic 6: STRINGED INSTRUMENTS and PITCH

CONCEPTS

1. Any musical instrument is a device for making sounds. In a stringed instrument (like a guitar, violin, or piano) a string is made to vibrate by plucking, bowing, or hitting. The vibrating string makes a soundboard or box vibrate, and this makes the air vibrate and causes the sound.
2. What musicians call a high note, or high pitch, is a rapid vibration (called a high *frequency* vibration). A low note, or low pitch, is a slow vibration (low frequency).
3. Two ways to make a string vibrate more rapidly (higher pitch) are to make it shorter or tighter. (You can also raise the pitch by replacing the string with a lighter one.)
4. Two ways to make a string vibrate more slowly (lower pitch) are to make it longer or looser. (You can also lower the pitch by replacing the string with a heavier one.)

STUDENT OBJECTIVES

1. List the names of stringed instruments and indicate how their strings are made to vibrate.
2. Recognize that there are two variables that determine how rapidly a string vibrates: its length and its tension. (Another is its weight, but we won't be studying this.)
3. Demonstrate that either tightening or shortening the string of a stringed instrument raises the pitch of the note that it plays.

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MATERIALS

- Pictures of stringed instruments
- One "scientist's guitar," which you should make beforehand (see pp. 36-37 for instructions). This requires one piece of lumber about 1" X 6" X 24", a small wood scrap about 1" X 1" X 6", a piece of corner bead (or even a ruler works well), two 4' lengths of fishing line between 15 and 20 lb. strength, five small nails, a pencil, two coffee cans, and some stones.
- If some of your students, parents, or colleagues play stringed instruments, invite them to bring their instruments.

Per group of students:

- A hardcover book
- A rubber band that will go tightly lengthwise around the book
- Two pencils

CLASSROOM ACTIVITIES

Good Vibrations

Students have made strings and rubber bands vibrate. Here they can become more familiar with instruments that use vibrating strings to make music.

PROCEDURE:

1. Divide students into small groups and have each group make a list of all the stringed instruments they know.
2. In a class discussion, have each small group name some of the stringed instruments they have on their list. Write them on the board. Ask students how each of these instruments is played. How are the strings made to vibrate? What else might be vibrating besides the strings?

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Explanation

The strings are made to vibrate in one of three ways: by plucking, bowing, or hitting. The strings then make the soundboard or box of each instrument vibrate. The large, flat vibrating surface of the board or box makes a rich, loud sound. Here is a list of some stringed instruments and the method used to play them.

Piano - hit
Violin - bow or pluck
Viola - bow or pluck
Cello - bow or pluck
Double Bass - bow or pluck
Guitar - pluck
Ukelele - pluck
Banjo - pluck
Dulcimer - pluck
Autoharp - pluck
Harpsichord - pluck
Clavichord - hit
Harp - pluck

3. Have the small groups decide how each of the instruments on their list is played. They can place a P, B, or H next to each instrument's name.

Book Banjos

Students will investigate the connection between the pitch of a stringed instrument and the length and tightness of its string.

PROCEDURE:

1. Make sure students know the difference between high notes and low notes. Introduce the term "pitch" as a way to describe this property of musical sounds.
2. Have the groups of students assemble their books, rubber bands, and two pencils to make a book banjo as they did with the topic "Sound Vibrations." Have them investigate ways to make the instrument's pitch vary by moving the pencils. How can they raise the pitch? Lower the pitch?

NOTE: During this activity students should focus on the part of the band between the two pencils and pluck just this part.

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Explanation

As the pencils are moved apart, the pitch will drop because the vibrating part of the band has been lengthened. As the pencils are moved together, the pitch will rise because the vibrating part of the band has been shortened.

3. Next see if the students can find out how the pitch varies if they make the string tighter. It is important here that they vary the tension but keep the length fixed. This is quite tricky. One way is to have one student grasp the rubber band at the two ends of the book and pull outward. A second student can hold the pencils so they don't move apart. A third is needed to pluck the rubber band. (They will find the tighter band produces a higher pitch.)
4. Have students share their findings. Make sure they are clear that there are two ways to raise pitch: by shortening or tightening the band. (Of course, you can also do both as is done in many instruments to achieve very high notes.) Unfortunately, students can not see how rapidly the band is vibrating (its frequency). Therefore, explain to them that shortening the string or tightening the string makes it vibrate more rapidly (high frequency). It is this more rapid vibration that we hear as higher pitch. Lengthening or loosening the string makes it vibrate more slowly (low frequency) and lowers the pitch.

The Scientist's Guitar

The "scientist's guitar" can be used to reinforce what students learned about pitch in the previous activity.

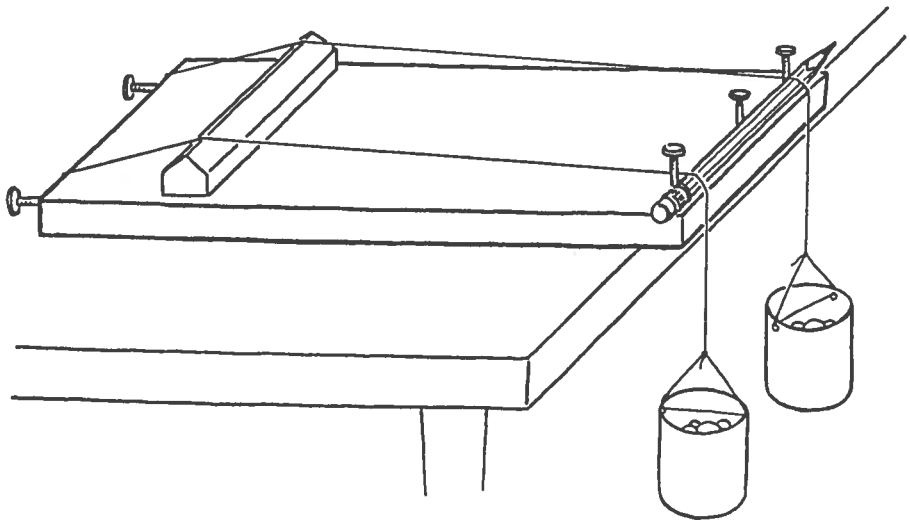
PROCEDURE:

1. Before conducting this activity, you need to have constructed the "scientist's guitar" as follows:

   Drive two nails into the end of the wooden board about 1/2" from each side. At the other end, drive three nails into the top as shown in the picture, so that a pencil can rest against them and just stick out beyond the end of the board. Use a nail to make two holes near the top of each coffee can, and fasten one end of a length of fishing line to each can as shown. Tie the other ends of the lines to the first two

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nails and pass the lines over the pencil so the cans hang freely side by side. Put a few stones into the cans to make the lines taut. Insert the small block of wood to form a bridge for both lines. The instrument will play better if you can plane or file the top of the block into the shape of a roof as shown. If you can't find a suitable block, a ruler on its edge works fairly well.



2. Group the whole class around the "guitar." Choose one of the strings to work with. Fill its coffee can about half full of stones and place the bridge at the opposite end of the board. Have one student pluck the string and have everyone listen to the note it plays. Now move the bridge a little toward the coffee can and pluck the same string again. Do this several more times and hear the note get steadily higher. If you practice this beforehand, you can find and mark the positions that give a major scale.

3. With one can full of stones and the other appreciably less full, pluck both strings and note how the tighter string plays the higher note. Repeat this with different loads of stones. If there's time, let some of your students try these experiments. Make the "scientist's guitar" available for students to experiment with.

4. If someone has brought a violin or other stringed instrument, have him/her show how shorter strings give higher notes and tighter strings do the same.

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Topic 7:

EAR HARPS

CONCEPT

1. All concepts from the previous topic, "Stringed Instruments and Pitch," apply.

STUDENT OBJECTIVE

1. Construct and play an ear harp.

MATERIALS

Per student:

- One piece of wood approximately 10" X 6" X 3/4" (The dimensions can vary quite a bit. Students may be able to bring in scraps of wood from home, or you may contact the shop teacher for some help in locating and cutting pieces.)
- Four screw eyes (at least 1-1/4" and preferably without any gap)
- Four nails (about 1"), and one larger nail for tuning
- 5' of fishing line between 10 and 20 lb. strength (The cheapest is fine, and colors are fun.)
- One ruler
- One pencil
- One large nail or screw to help turn the screw eyes on the harps

Per class:

- A few adult assistants to help tie strings, especially with younger students
- At least ten hammers
- Several scissors
- One or two pliers for tightening gaps in the screw eyes

NOTE: If you are having difficulty finding or affording enough materials for each student, you could have the students work in groups.

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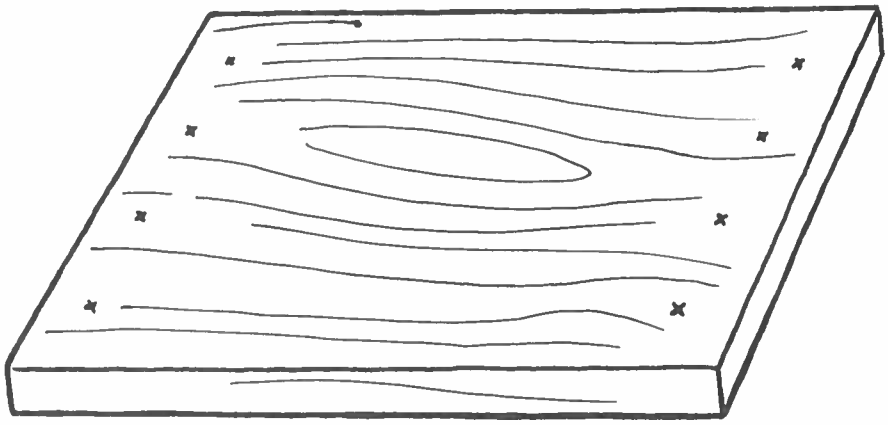
Classroom Activities

Constructing an Ear Harp

This activity gives students a chance to build their own stringed instruments, which they can tune and play.

Procedure:

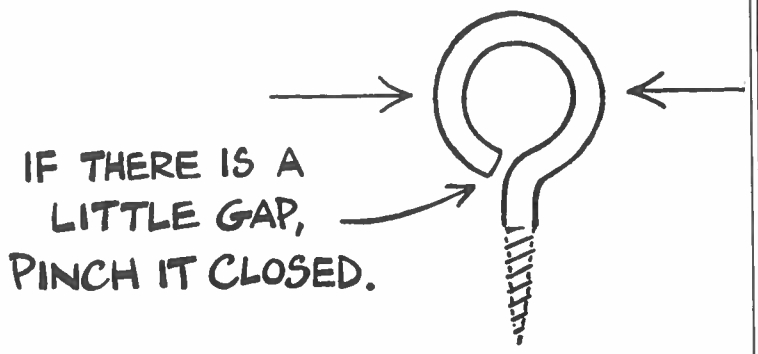
1. To make a four-stringed ear harp, the students must make four marks equally spaced across each end of the wood about 1/2" from the end. For example, if the wood is 6" wide, you could make the marks at 1-1/2", 2-1/2", 3-1/2", and 4-1/2". Have them repeat this procedure along the other end of the wood.



NOTE: Whatever the size of wood, make sure that the four marks are placed at least 1" apart. Younger students may need more guidance. Older students may want to do this whole project on their own.

2. Use a nail to pound small starter holes at each mark along one end of the wood. Then screw one screw eye in each mark so that a few threads of the screw are still showing.

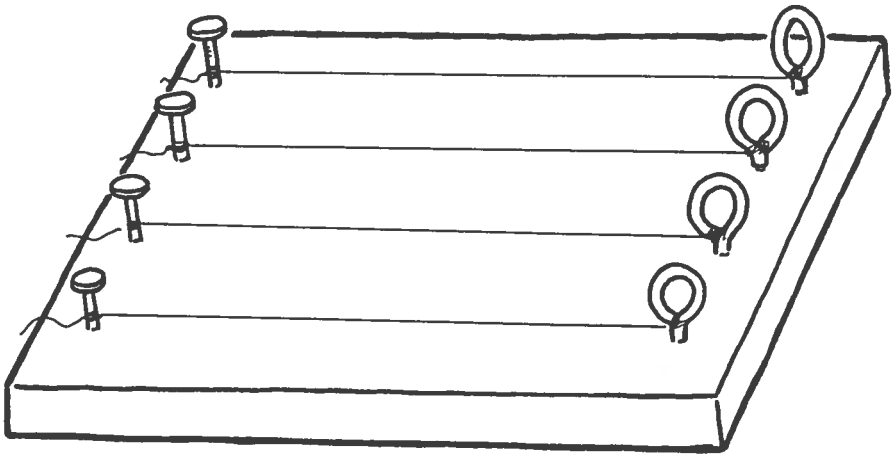
3. Once the screw eyes are in place, have the students use the pliers to pinch any gaps closed. Otherwise the strings may slip through the gaps.



4. Pound a nail in each mark at the other end of the wood.

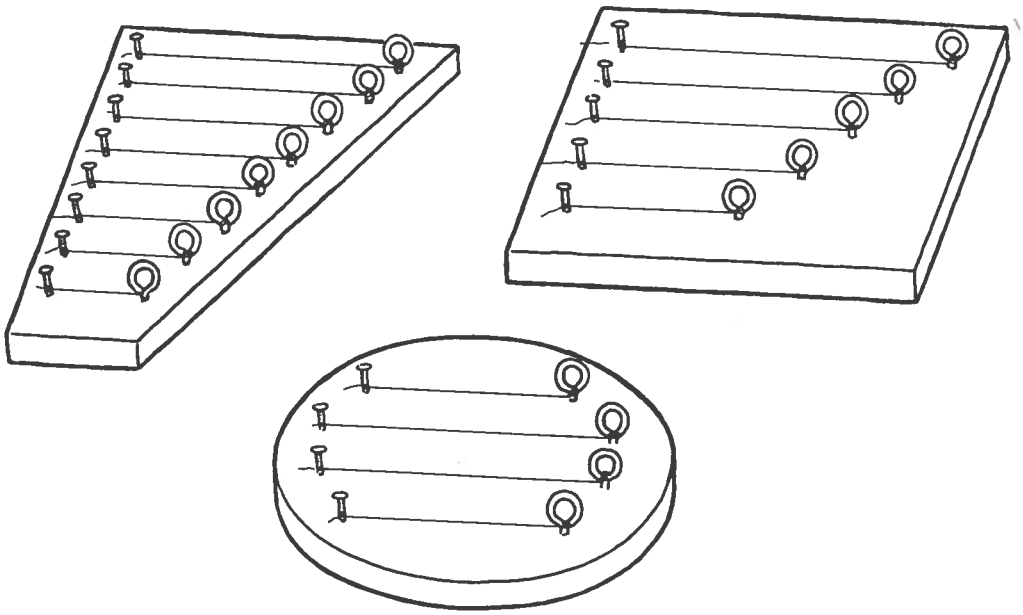
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5. Cut the fishing line into four equal pieces. Tie one end of each piece through one of the screw eyes. Allow the knot to slip to the bottom of the eye. Pull the string tight to the opposite nail, wrap it several times around, and then tie it. This completes the ear harp.



6. Have students place the harp next to their ears with the strings facing out. They can then pluck the strings and hear the notes quite clearly.

NOTE: Students who want a more challenging project could make an ear harp with strings of varying lengths or with more strings. With eight strings they can tune it to a whole scale.



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Tuning the Ear Harps

Using what they have learned, the students can now tune their ear harps.

PROCEDURE:

1. Have the students use nails or screws to twist the screw eyes on their ear harps. This lets them adjust the tension in the strings and hence the pitch of the notes. Some students may be satisfied with just making the various strings play different notes, others may want to make them play a scale or arpeggio. (An eight-string harp can play a scale; a four-string harp an arpeggio.) If all the strings are the same length, it makes no difference which they make high and which low. If some strings are longer than others, it is natural and easier to make the longer strings be the lower ones.

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Topic 8:

SOUND WORDS

(Optional)

Student Objectives

1. Make lists of words that pertain to sound.
2. Write a paragraph, story, or poem using some of these words.
3. Catch up on unfinished projects.

Classroom Activity

The Melodious Scraping Sound

This writing exercise will help students broaden their vocabulary pertaining to sound.

PROCEDURE:

1. Have a brainstorming session where students list words related to sound (e.g., bang, soft, scraping, vibrate, growl, piercing, frequency, trombone, soothing, melodious). They might want to use a thesaurus to assist them.
2. You might have students categorize their words according to the parts of speech (nouns, verbs, adjectives, etc.).
3. If you can obtain a copy of Edgar Allen Poe's *The Bells*, read it to the class. It is a wonderful example of "sound words."
4. Have students write a paragraph, story, or poem using as many of their words as is appropriate.

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PHYSICS for FUN

Topic 9: WIND INSTRUMENTS

SPECIAL NOTE: **This and the following topic ("Sound Sensations") are all about making sounds by blowing across bottles. Younger students, especially third graders, may find this difficult. Additional adult guidance may help, and you might have the students practice this skill in advance.**

CONCEPTS

1. In a stringed instrument, the sound originates with the vibration of a string; in a wind instrument, like a trumpet or flute, it originates with the vibration of air inside a tube. Because the tube has at least one opening, the vibrating air inside the tube causes the air outside to vibrate, and this vibration is the sound that travels from the instrument outward to the listeners.
2. The longer the tube, the more slowly the air inside vibrates. Thus long tubes, like trombones and tubas, make low notes. Short tubes, like whistles and piccolos, make high notes.

STUDENT OBJECTIVES

1. Experience the different pitches of notes produced by blowing across bottles of varying sizes and containing varying amounts of water.
2. Tune a set of bottles to a major scale by adding water to bottles of similar size.

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MATERIALS

Per student (to bring from home):

- One bottle of any shape or size (Suggest that they look for odd bottles so you don't end up with all of one kind. Have them soak off the labels and rinse out the bottles.)
- One soda bottle (10, 12, or 16 fluid ounces) with the label soaked off (It would be simpler if all of these were the same kind. These can often be obtained from stores that recycle.)

Per class:

- Wind instruments played by students, parents, or colleagues
- Masking tape
- Pencils
- Rulers marked in centimeters
- Several pitchers or large bottles of water
- Sponges or paper towels to wipe up spills
- Several lengths of clear, flexible plastic tubing 2' to 3' long (Tubing with an inner diameter of 3/8" to 1/2" works well.)
- Rubbing alcohol and cotton balls or paper towels for sterilizing the mouths of bottles and tubing
- Piano, keyboards, xylophones, or other instruments to help students tune the bottles

CLASSROOM ACTIVITIES

Musical Bottles

Learn, by blowing across the tops of differently shaped bottles, that tall bottles produce low notes and short bottles produce high notes.

PROCEDURE:

1. Have students blow across the tops of various bottles to make a sound. Some will be able to do this easily and some will not. If there are any children who play the flute, they may have some pointers for the rest of the class.
2. Once they have mastered the technique of blowing, have students compare the sound of their bottle to their neighbor's. Which is higher, which is lower? Can they explain why the pitches are different? What do they think is vibrating to produce the notes they are playing? Do they see the connection to what they learned about pitch and stringed instruments with the topic "Stringed Instruments and Pitch"?

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3. Find the highest and the lowest notes in the class. Compare the sizes of these bottles.

Explanation

We learned with the topic "Stringed Instruments and Pitch" that the pitch of a note varies with the length of the object vibrating. In stringed instruments, it is the string that is vibrating. In a wind instrument there is no string. What is vibrating is the column of air inside the instrument. This, in turn, causes the air near the openings of the instrument to vibrate. These vibrations travel outward and we eventually hear the note. The pitch is different according to the length of the column of air inside the tube or instrument. The longer the tube, the lower the note, just as with a string. (The width of the column also affects the pitch somewhat. Don't be surprised if two different bottles of the same length produce somewhat different notes.)

4. Have students or adults who play a wind instrument demonstrate their instrument and explain how it works. Where is the column of air? Where are the holes, and how do they play the instrument in order to change the pitch?

NOTE: Be aware that the ways wind instruments work are diverse and complicated. A trombone changes its length by sliding one tube in and out of another, thus changing its overall length. Trumpets and horns change their length by opening and closing valves. Woodwinds (flutes, clarinets, etc.) change their length by opening and closing small holes. All wind instruments can also change their pitch by blowing in different ways to produce what are called harmonics.

Water Trombone

Using a flexible plastic tube, partially filled with water, demonstrate how changing the length of the column of air changes the pitch.

PROCEDURE:

1. Have students work in groups. The number of groups is determined by the number of plastic tubes you have.
2. Each group should have a pitcher of water and a length of tubing. Have students carefully pour water into the plastic tube so that it is

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about half full. There should be no air bubbles in the column of water. One by one, have students blow across one end of the tube while holding the other end to keep the water from flowing out. While continuing to blow, move the other end of the tube up and down so that the water level varies. The note should slide up and down in pitch depending on the length of the column of air.



3. Ask students what is happening here. Why is the note changing pitch? What instrument is similar to this? If a slide trombone has not already been demonstrated, try to get one to show the students how it works.

Pop Bottle Ensemble

Students can alter the pitch of their bottles by putting water in them since the water changes the length of the air column in the bottle. By tuning bottles with water, students will be able to produce at least five notes of a major scale, and the class can play the scale.

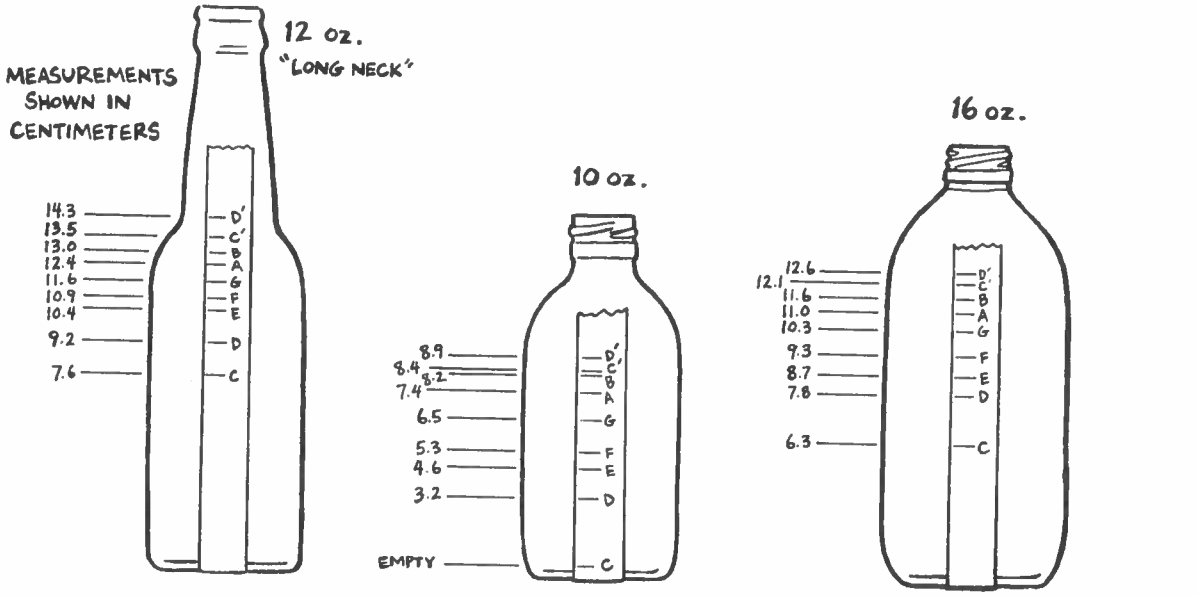
PROCEDURE:

1. Have each student place a piece of masking tape down the side of his/her 10-, 12-, or 16-ounce bottle.
2. Divide the class into five groups. Assign each group a note: Middle C, D, E, F, or G.
3. Give each group the appropriate measurements for their note and their size bottle (see illustrations on p. 49). Have them measure up from the bottom of the bottle by placing their ruler vertically next to the bottle. Some rulers do not start the measurement markings at the end of the ruler. If this is the case, have students place their bottles at the edge of the table and line up the first mark of the ruler with the table's edge. Students should be at eye level when reading the measurements. Have students make a mark on the tape at the appropriate place and then write the name of their note beside the mark.

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4. Have students fill their bottles with water up to their pencil marks and practice playing their note. Once they can play the note clearly, have them check the note against the piano or other instrument. Encourage group work in deciding when a note is in tune. Some students will have a better or more trained musical ear than others and can hear the differences more easily.

NOTE: Encourage students to hold bottles upright when they blow because the pitch can vary if they tip the bottle. Even similar bottles filled to the same depth may produce somewhat different notes, depending on the type of bottle, the way an individual blows, and the way the measurement is taken. You may need to do some fine tuning by adding or removing a little water. Don't worry about getting it too precise.



ALL MEASUREMENTS ARE ONLY APPROXIMATE AND VARY FROM BOTTLE TO BOTTLE.

5. Collect all the bottles and store them to use with the following topic, "Sound Sensations."

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PHYSICS for FUN

Topic 10:

SOUND SENSATIONS

CONCEPT

1. All concepts from the previous topic, "Wind Instruments," apply.

STUDENT OBJECTIVE

1. Play tunes with bottles.

MATERIALS

- The pop bottles tuned for the previous topic "Wind Instruments."
- A copy of the "Conductor's Score," included in this lesson plan (p. 53), to be used as an overhead or copied onto the board

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Classroom Activity

Grand Finale

You, or a student, will conduct the whole class playing tunes on their musical bottles.

Procedure:

1. Have each student reclaim his or her bottle from the previous topic's bottle tuning. Make sure students are aware of the notes their bottles play. If necessary, let students first practice playing notes with their bottles.

2. You may now conduct the whole class in playing some exercises. For example, you may write the following exercises on the board:

   C D E F G F E D C C D E F G G F E D C C E D F E G G G E F D E C

   As you point to each note, all the students assigned to that note should play it.

3. Now conduct the class in some of the five-note tunes in the "Conductor's Score." Write the tunes on the board or use an overhead projector. Notice that the score gives no indication of the rhythm, but the tunes are all well known, so you will be able to point at the notes in the correct rhythm.

4. You may want to have students try conducting, and you could divide the class into smaller groups, each with its own conductor.

5. The more ambitious may want to retune some of their bottles to play more notes of the scale, so they can play some of the "more ambitious tunes" in the "Conductor's Score."

NOTE: High C and High D are labeled C' and D' in the score.

6. Your students may want to perform for each other, another class, or their parents.

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Conductor's Score

Five-Note Tunes

Lightly Row (notes C through G)

G E E F D D C D E F G G G G E E F D D C E G G C

When The Saints Come Marching In (notes C through G)

C E F G C E F G C E F G E C E D

E E D C C E G G G F E F G E C D C

Ode To Joy / Beethoven's 9th (notes C through G)

E E F G G F E D C C D E E D D E E F G G F E D C C D E D C C

Mary Had A Little Lamb (notes C, D, E, G)

E D C D E E E D D D E G G E D C D E E E E D D E D C

Some More Ambitious Tunes

Twinkle Twinkle Little Star (notes C through A)

C C G G A A G F F E E D D C G G F F E E D G G F F E E D

C C G G A A G F F E E D D C

Oh Susanna (notes C through A)

C D E G G A G E C D E E D C D C D E G G A G E C D E E D D C

F F A A G G E C D C D E G G A G E C D E E D D C

Three Blind Mice (notes C through C')

E D C E D C G F F E G F F E G C' C' B A B C' G G

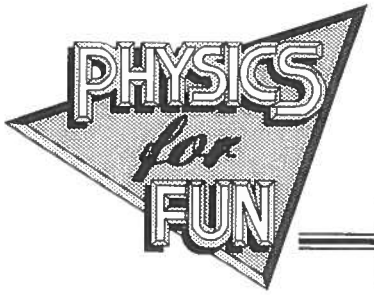
G C' C' B A B C' G G G G C' C' B A B C' G G F E D C

Swanee River (notes C through D)

E D C E D C C' A C' G E C D E D C E D C C' A C' G E C D E D C

B C' D' G G A G C' C' A F A G E D C E D C C' A C' G E C D E D C

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APPENDIX

Name: _____ Date: _____

Sound Unit Test

Answer as completely as you can. Write all that you know.

1. What produces **sound**?

_____

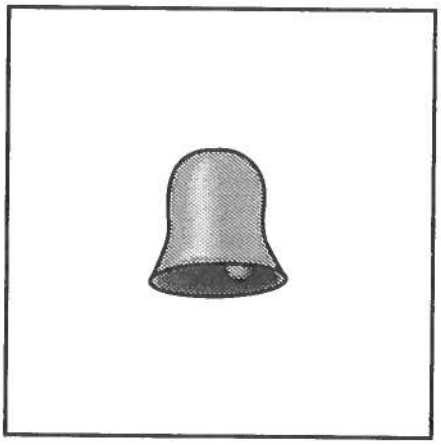
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2. How does **sound** travel?

_____

_____

3. Draw on the picture how **sound travels** through a room when a bell rings.



Explain your drawing with words.

_____

_____

_____

_____

_____

_____

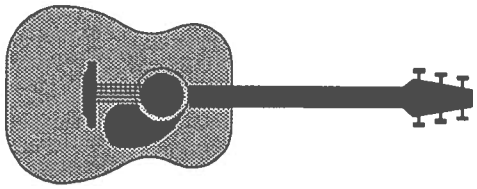
4. How do your **ears** hear sound?

_____

_____

_____

5. Explain how this nonelectric guitar makes and **amplifies** sound. ("**Amplifies**" means makes louder.)



6. If you want to make a **higher pitched sound** on the guitar, would you tighten or loosen its strings? Check (✓) one answer.

   ❑ Tighten

   ❑ Loosen

   Why did you choose your answer?

7. Give one good reason why you cannot hear sound in **outer space**.

8. Explain a way you could **measure** the **speed** of sound.

SCORING KEY

Sound Unit Test

Answer as completely as you can. Write all that you know.

(2 points possible)

1. What produces **sound**?

Sound is a (1) vibration traveling through a (1) medium.

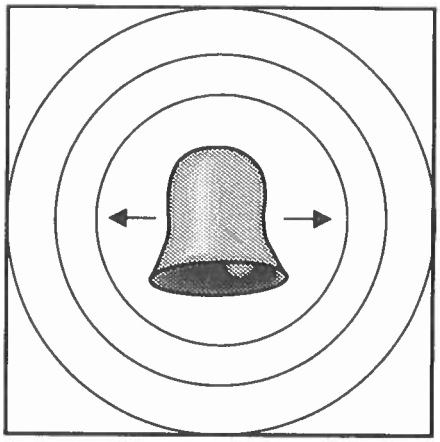
2. How does sound **travel**?

(2 points possible)

Sound travels in (1) compression waves which act like (1) dominoes tipping each other over.

(4 points possible)

3. Draw on the picture how **sound travels** through a room when a bell rings. (1)



Explain your drawing with words.

Sound (1) travels in all directions from its source and (1) fills up the whole room (1) (but it is louder near its source).

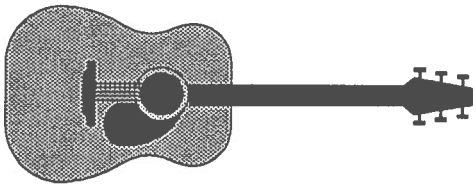
(4 points possible)

4. How do your **ears** hear sound?

(1) Vibrations hit your ears. (1) Parts of the ears (1) vibrate and send (1) messages to the brain via nerves.

5. Explain how this nonelectric guitar makes and **amplifies** sound. (**"Amplifies"** means makes louder.)

(4 points possible)



(1) The player plucks the strings.

(1) The strings vibrate.

(1) The vibrations enter the box.

(1) The box makes sound bounce around so it grows louder.

6. If you want to make a **higher pitched sound** on the guitar, would you tighten or loosen its strings? Check (✓) one answer.

(3 points possible)

(1) ☑ Tighten

☐ Loosen

Why did you choose your answer? (1) A tighter string has a higher frequency.

(1) A higher frequency has a higher pitch.

7. Give one good reason why you cannot hear sound in **outer space**.

(2 points possible)

(1) There is no air, which (1) means no medium for the sound to travel through.

8. Explain a way you could **measure** the **speed** of sound.

(4 points possible)

There are various answers, but one possibility follows:

(1) Stand by a large object and make a sound.

(1) Time how long it takes for the echo to travel the (1) measured distance.

(1) Divide the distance by the time.

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